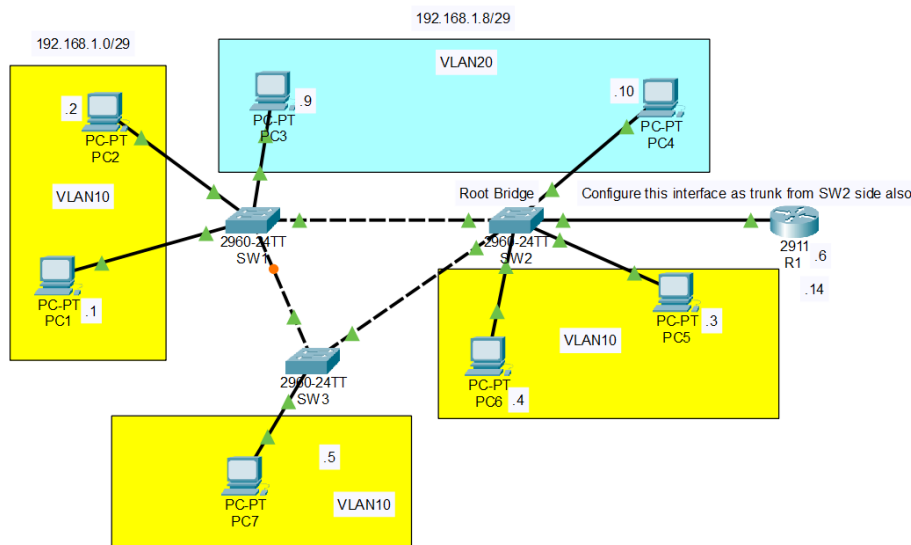


Objective – To implement and test VTP and EtherChannel of the Layer 2 in Cisco Packet Tracer.

Network Topology:



1. VLAN Trunking Protocol (VTP):

First attempt failed

Reason – I configured VTP modes, but not the same VTP Domain.

Now using the command `<#vtp domain cisco>` on SW1, SW2 and SW3.

Keeping SW1 in `<vtp mode transparent>`

SW2 in `<vtp mode server>`

SW3 in `<vtp mode client>`

Results

SW2:

```
SW2(config)#vlan 10
SW2(config-vlan)#name Engg
SW2(config-vlan)#exit
^
% Invalid input detected at '^' marker.

SW2(config-vlan)#exit
SW2(config)#vlan 20
SW2(config-vlan)#name HR
SW2(config-vlan)#exit
SW2(config)#do sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/2
10	Engg	active	Fa0/2, Fa0/4
20	HR	active	Fa0/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW2(config)#

SW3:

```
SW3(config)#do sh vlan br
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2 Fa0/4
10 VLAN0010	active	
20 VLAN0020	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW3(config)#do sh vlan br
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2 Fa0/4
10 Engg	active	
20 HR	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW3(config)#
```

SW1:

```
SW1(config)#vtp domain cisco
Changing VTP domain name from NULL to cisco
SW1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to cisco.

SW1(config)#do sh vlan br
```

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10 VLAN0010	active	Fa0/2, Fa0/3
20 VLAN0020	active	Fa0/1
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW1(config)#
```

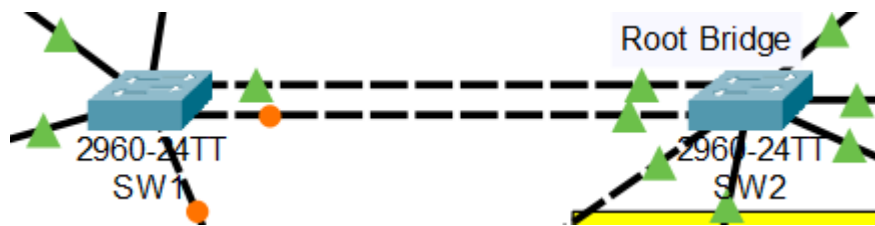
No change, as SW1 is in transparent mode. A switch in transparent mode neither updates its own VLAN database with the server nor shares its own changes with other switches. It just forwards the server VTP database changes to client VTP data changes. Client follows them, as we can see in SW3 CLI.

2. EtherChannel:

By subnetting, we know there is a chance to add 3 more end devices to VLAN 20. While dealing with real-world problems, that number would be in hundreds or even thousands. More bandwidth would be needed to tackle this problem, cause a single link between SW1 and root bridge SW2 won't be enough with its limited bandwidth. As network traffic increases, a single link between switches can become a bottleneck.

Solution 1 – Adding one more physical link between SW1 and SW2 for load balancing.

But that link would be blocked to avoid layer 2 loops, as we have our Spanning Tree Protocol.

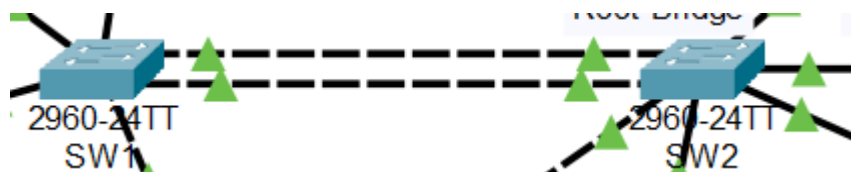


Solution 2 – Make the different physical links logically one for the switches to avoid STP blocking. And here comes the concept of EtherChannel.

Command to logically make EtherChannel of two physical links:

(config-if)#channel-group 1 mode active (LACP)

Note – Don't forget to configure the newer physical link exactly same as the older one with which it is being paired for EtherChannel. Things like access or trunk, VLAN numbers, interfaces, native VLANs, allowed VLANs, VLANs in management domain, active VLANs, etc, all should be the same.



SW1:

```
SW1#sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby  (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	LACP	Fa0/4 (P) Fa0/6 (P)

SW1#

SW2:

```

SW2(config)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

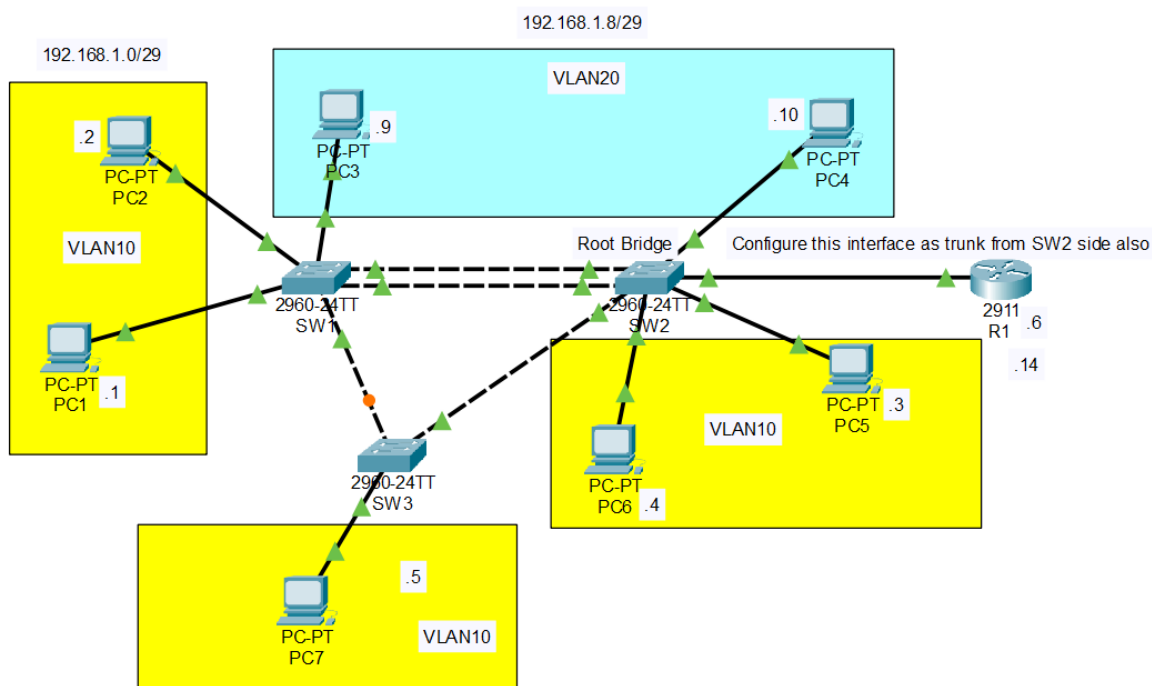
Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)                LACP      Fa0/3(P) Fa0/6(P)
SW2(config)#

```

The capital “P” mode is really crucial.

This indicates that both the interfaces have been grouped into 1 EtherChannel and treated as one logical link. STP blocking won’t be here, and this link will be used for load balancing between both switches.

Final topology:



An important image from the PC1 command prompt:

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.9

Pinging 192.168.1.9 with 32 bytes of data:

Reply from 192.168.1.9: bytes=32 time<1ms TTL=127
Reply from 192.168.1.9: bytes=32 time<1ms TTL=127
Reply from 192.168.1.9: bytes=32 time<1ms TTL=127
Reply from 192.168.1.9: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```